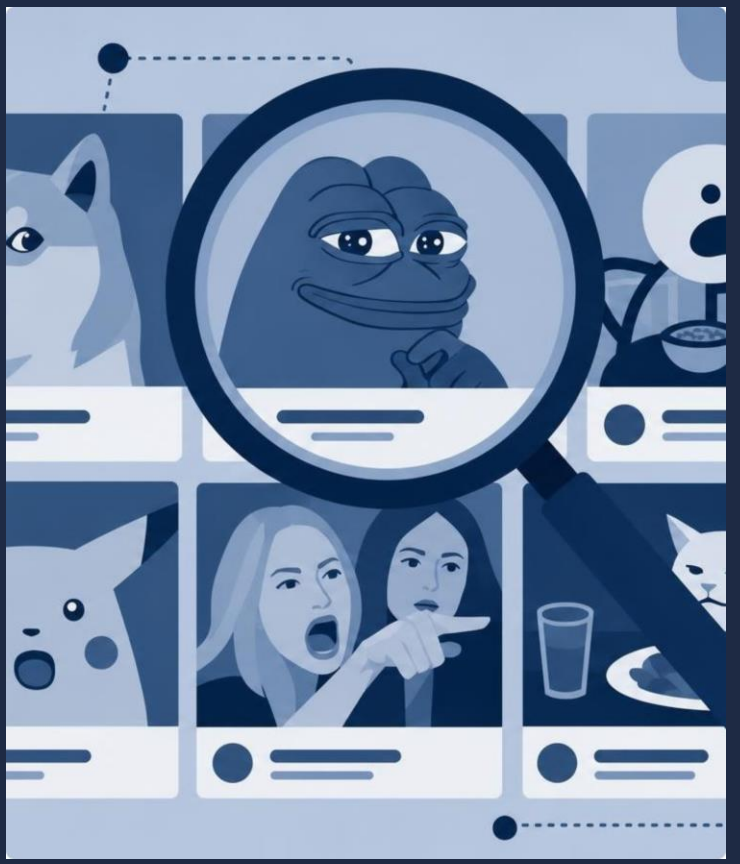
Beyond Vision: Multimodal Meme Search & Reaction Suggestion



Moving from literal pixel matching to hidden human sentiment and visual metaphor

Group: Beyond Vision | Course: COMP5523

Presentation Roadmap



01

Domain Challenges

02

Dataset & 3 Models



03

Live Demo (1 Min)

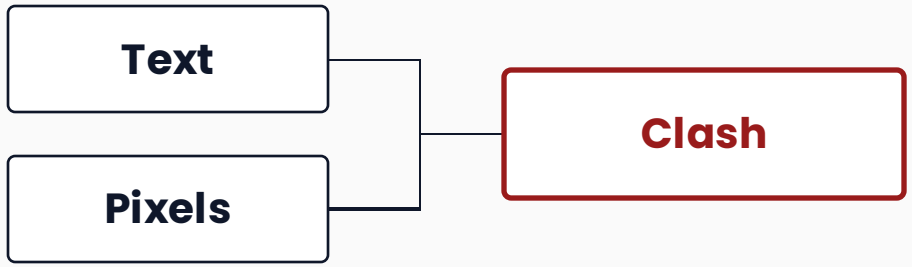
04

Anti-Leakage Eval

05

Results & Ablation

Domain Challenges: How to Retrieve a Meme?



Elementary Vision

Literal pixel searching misses the joke

Typographic intrusion: overlaid text hijacks attention

Visual contradictions between image and caption

BEYOND VISION

Search by Sentiment:

abstract emotion → reaction meme

Search by Metaphor:

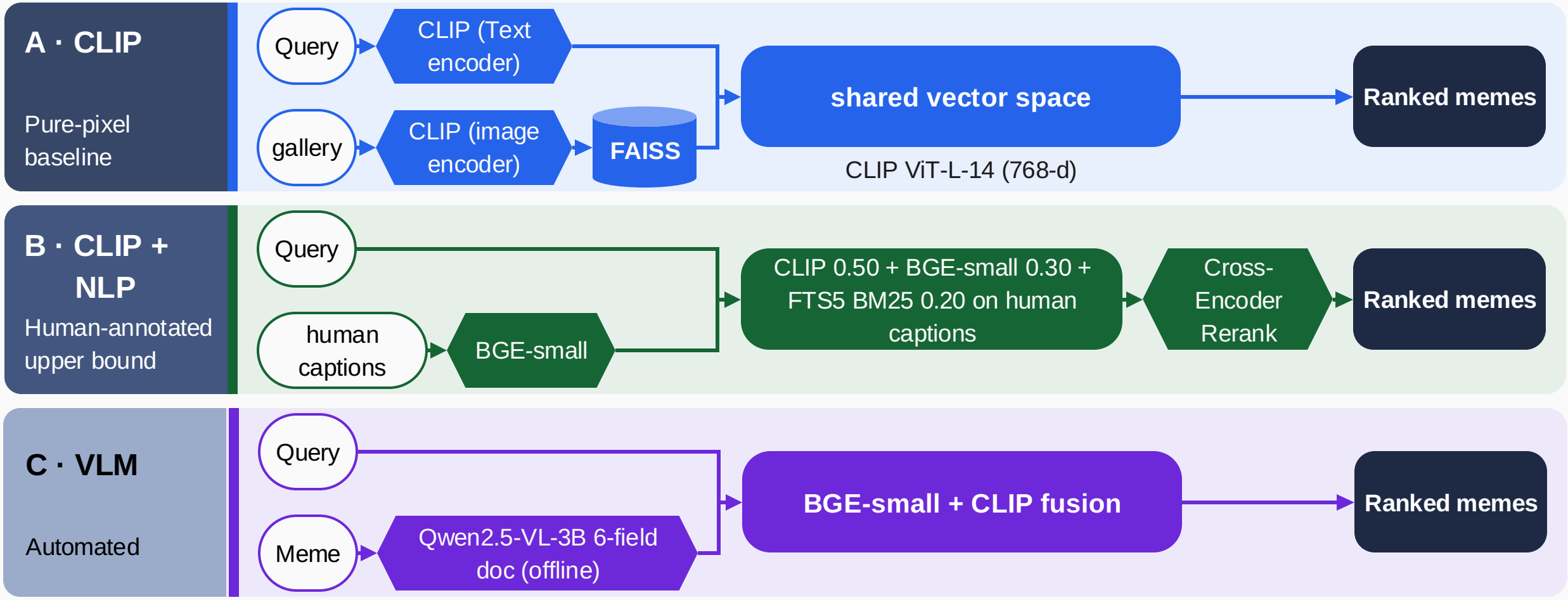
literal objects ← → implied meanings

Key objective: benchmark pure pixels vs automated VLM vs human-annotated upper bound

Dataset & 3 Model Structures

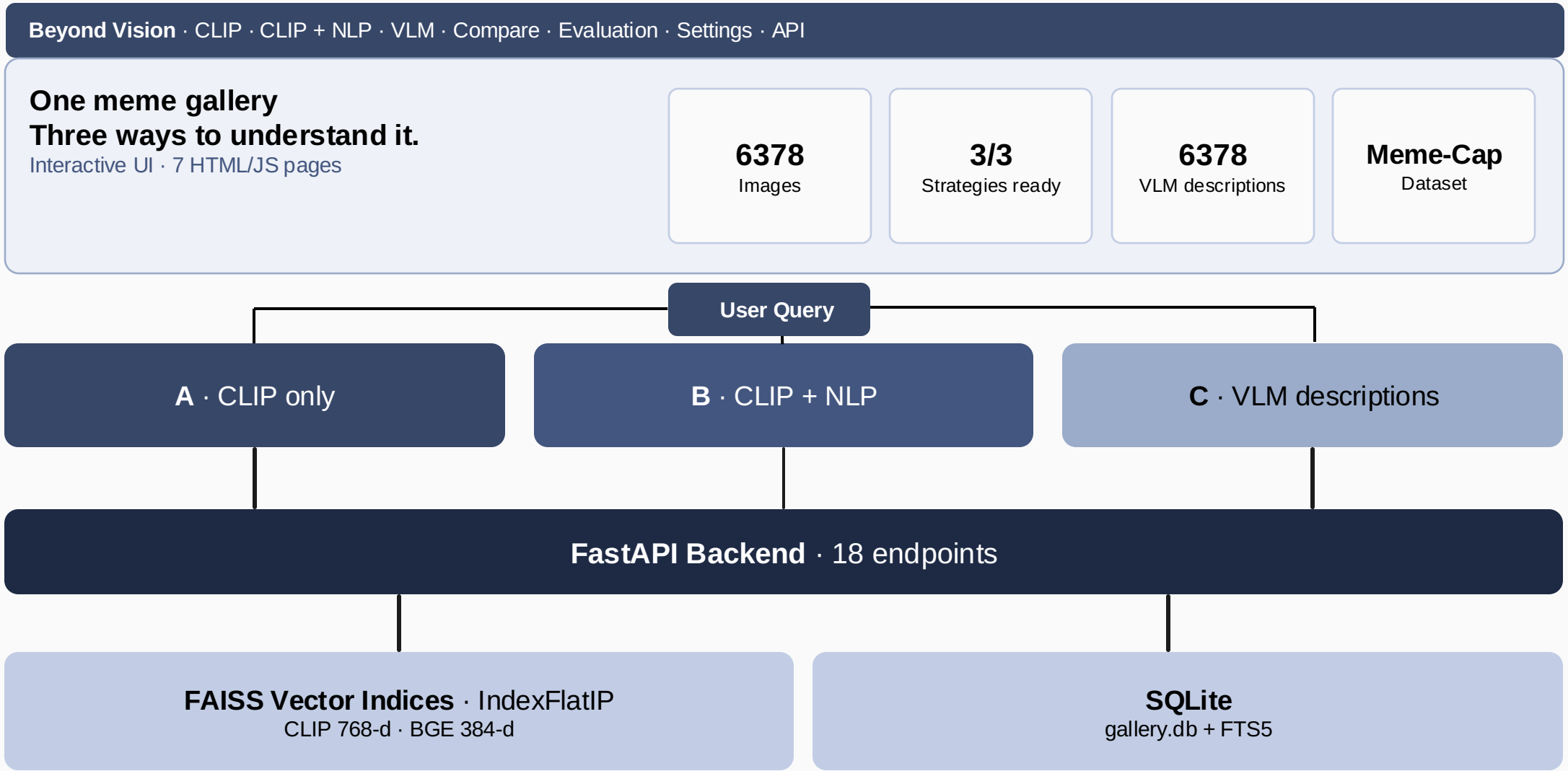
MemeCap (EMNLP 2023): 6,378 verified memes (5,820 trainval + 558 test) · 1,968 noisy entries stripped

Data fields: Image file + Reddit title + human annotations (scene & meme caption + metaphors pair

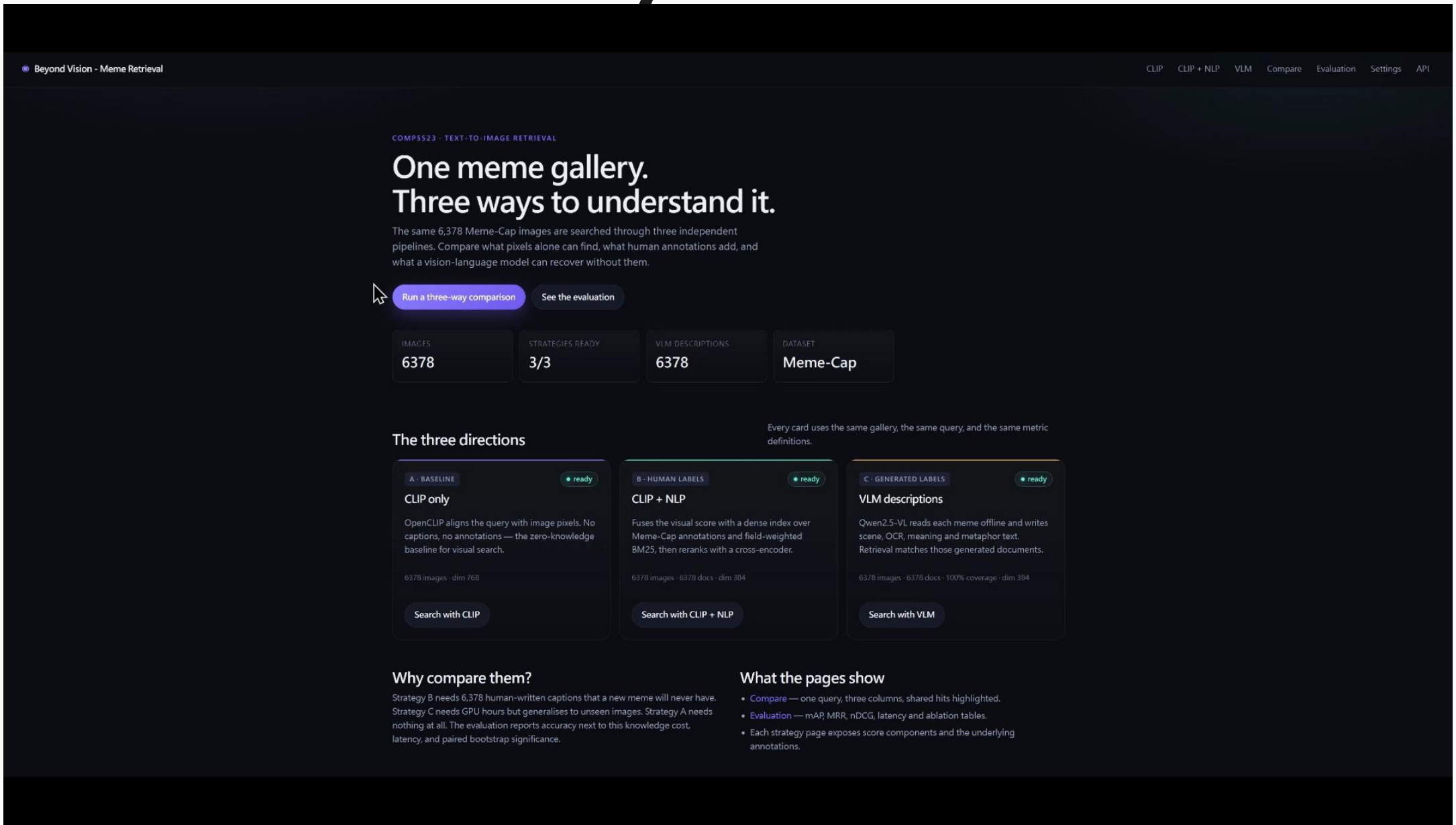


6 fields: scene · OCR · meaning · metaphors · emotions · keywords

System Architecture & Implementation



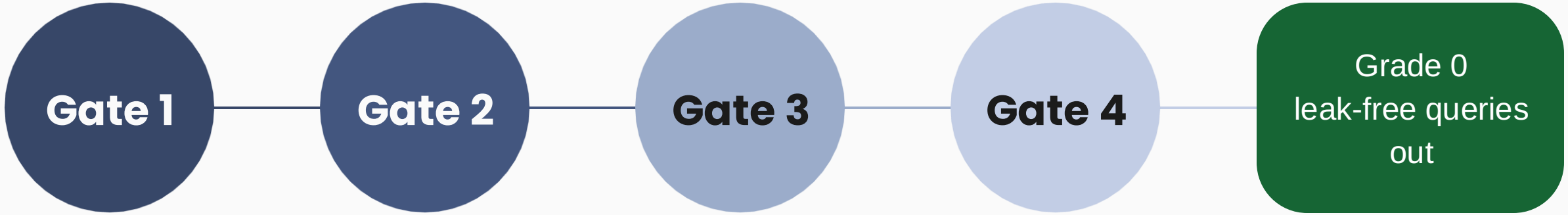
1-Minute Live System Demonstration



Anti-Leakage Evaluation Dataset

240 queries from 558 unseen test memes · **80 Visual · 80 Intent · 80 Metaphor**

Deterministic Graded Ground Truth:
Grade 2 (source test meme), Grade 1 (near-equivalent memes or shared rare metaphor), Grade 0 (all others)



Overlap

8-gram overlap check

Length

Short-caption gate

Copy run

Max contiguous copy run <5 tokens

Jaccard

Token Jaccard <0.60

Evaluation Results: Outcomes & Model Comparison

Strategy	Human Labels	Overall mAP	Visual (80)	Sentiment (80)	Metaphor (80)	Median Latency (p50)
A. CLIP (pure pixels)	No	**0.4395**	0.315	0.444 (worst)	0.559	14 ms (fastest)
C. VLM (0.2 CLIP)	No	**0.4519**	0.247 (worst)	0.539	0.570	35 ms
B. CLIP + NLP (Upper bound)	Yes (Oracle)	**0.7608 (best)**	0.631	0.739	0.912	226 ms *(43 ms no-rerank)*

Best Overall:

B · CLIP + NLP, mAP = 0.7608

Sets the performance ceiling when human crowd-sourced explanations and cross-encoder reranking are available, at the cost of a 226 ms latency.

Best Annotation-Free Model:

C · VLM, mAP = 0.4519

Beats pure CLIP (0.4395) while running in 35 ms. Most notably, on "Beyond Vision" **Intent** queries, Strategy C jumps from Strategy A's worst score of **0.444 up to 0.539 (+9.5%)**

Worst Category Score:

C · VLM on Visual = 0.247

Default Strategy C loses to pure CLIP (0.315) on literal visual queries until pixel weight is raised to 0.35.

Ablation Insights & Root-Cause Diagnosis

ABLATION STUDY OUTCOMES

Strategy B without Cross-Encoder Reranker

0.7608 → 0.6887 mAP · BGE -reranker-base is the largest accuracy contributor

−0.0721 mAP

B without CLIP Pixels (weights.clip = 0.0)

0.7608 → 0.7802 mAP

+0.0194 mAP

C without CLIP Pixels (clip_weight = 0.0)

0.4519 → 0.3922 mAP

−0.0597 mAP

ROOT-CAUSE DIAGNOSIS

1. Why BGE is not to blame for Strategy C's visual drop

Strategies B and C use the exact same BGE-small (384-d) text embedder. Holding BGE constant proves the bottleneck is Qwen2.5-VL-3B's image-to-text translation.

2. Why pixels hurt B (−0.0194) but rescue C (+0.0597)

Strategy B: human annotators already describe scene and lore accurately, so noisy CLIP pixels dilute clean text.

Strategy C: Qwen2.5-VL-3B is a lossy bottleneck: explains the joke (0.539 Intent) but drops details (0.247 Visual).

Work Distribution

Team – High level organization

- Idea initiatives
- Feasibility assessment
- Examine AI reasoning
- System testing
- Benchmark verification
- Anti-leakage auditing

GenAI – Implementation

- Coding



Thank you!